Problem Set 30

The Fundamental Theorem of Calculus, Part 2

1

(a) Find the following indefinite integrals, and check your answers by differentiating them.

i. $\int \cos x\, dx$

Solution

$\boxed{\sin x + C}$. (The derivative of $\sin x + C$ is $\cos x$.)

ii. $\int u^3\, du$

Solution

$\boxed{\frac{1}{4}u^4 + C}$. (The derivative of $\frac{1}{4}u^4 + C$ is u^3.)

iii. $\int \frac{x}{\sqrt[3]{x}}\, dx$

Solution

We can rewrite this as $\int x^{2/3}\, dx$, which is $\boxed{\frac{3}{5}x^{5/3} + C}$. (The derivative of $\frac{3}{5}x^{5/3} + C$ is $x^{2/3}$.)

iv. $\int e^{-t}\, dt$

Solution

$\boxed{-e^{-t} + C}$. (The derivative of $-e^{-t} + C$ is e^{-t}.)

(b) Find the antiderivative $F(x)$ of $4 - \dfrac{3}{1+x^2}$ which satisfies $F(1) = 0$.

Solution

The antiderivatives of $4 - \dfrac{3}{1+x^2}$ are $4x - 3\arctan x + C$. We want to find the value of C which makes it so that $F(1) = 0$:

$$\begin{aligned} 4 - 3\arctan 1 + C &= 0 \\ 4 - \frac{3\pi}{4} + C &= 0 \\ C &= \frac{3\pi}{4} - 4 \end{aligned}$$

So, $F(x) = \boxed{4x - 3\arctan x + \frac{3\pi}{4} - 4}$.

2 In February 2012, several people on campus got norovirus. Suppose $r(t) = 6 - 3\sqrt{t}$ gave the rate of change of the number of people on campus with norovirus, in people per day, where t is measured in days and $t = 0$ represents February 10. On February 14, 100 people on campus had norovirus.

(a) How many people on campus had norovirus on February 19?

Solution

The number of people with norovirus on February 19 was $100 + \int_4^9 (6 - 3\sqrt{t})\,dt$. One antiderivative of $6 - 3\sqrt{t} = 6 - 3t^{1/2}$ is $6t - 2t^{3/2}$, so

$$\begin{aligned} 100 + \int_4^9 (6 - 3\sqrt{t})\,dt &= 100 + \left(6t - 2t^{3/2}\Big|_4^9\right) \\ &= 100 + ([54 - 2 \cdot 27] - [24 - 2 \cdot 8]) \\ &= \boxed{92} \end{aligned}$$

(b) How many people on campus had norovirus on February 11?

Solution

The number of people with norovirus on February 11 was

$$\begin{aligned} 100 + \int_4^1 (6 - 3\sqrt{t})\,dt &= 100 + \left(6t - 2t^{3/2}\Big|_4^1\right) \\ &= 100 + ([6 - 2 \cdot 1] - [24 - 2 \cdot 8]) \\ &= \boxed{96} \end{aligned}$$

(c) At what times between February 10 and February 20 was the number of people with norovirus increasing? At what times was it decreasing?

Solution

We can answer this by figuring out when $r(t) = 6 - 3\sqrt{t}$ is positive or negative; that is, we want to make a sign chart for $r(t)$. So, we should first figure out when $r(t)$ is 0:

$$\begin{aligned} 6 - 3\sqrt{t} &= 0 \\ 6 &= 3\sqrt{t} \\ 2 &= \sqrt{t} \\ 4 &= t \end{aligned}$$

Then, we can test a value less than 4 and a value greater than 4; for example, $r(1) = 6 - 3 > 0$ and $r(9) = 6 - 3 \cdot 3 < 0$. That gives us the following sign chart for $r(t)$:

		4	
sign of r	$+$		$-$

So, the number of people is increasing until February 14 and decreasing after that.

(d) Are your answers consistent with each other and with the fact that 100 people had norovirus on February 14?

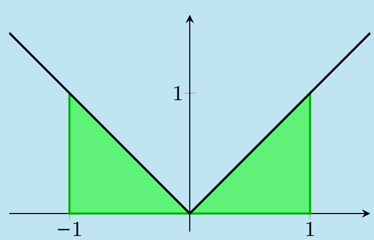
Solution

Yes, February 14 was the peak, so the number of cases on February 11 and 19 should both be lower than 100.

3 (a) Find $\int_{-1}^{1} |x|\, dx$ by interpreting it as a signed area.

Solution

We can picture this integral as the signed area under $y = |x|$ from $x = -1$ to $x = 1$:



This area is $\boxed{1}$.

(b) Show that $\frac{|x|^2}{2}$ is *not* an antiderivative of $|x|$ by showing that $\int_{-1}^{1} |x|\, dx \neq \left.\frac{|x|^2}{2}\right|_{-1}^{1}$.

Solution

We just found that $\int_{-1}^{1} |x|\, dx = 1$. On the other hand, $\left.\frac{|x|^2}{2}\right|_{-1}^{1} = \frac{1}{2} - \frac{1}{2} = 0$, so $\int_{-1}^{1} |x|\, dx \neq \left.\frac{|x|^2}{2}\right|_{-1}^{1}$.

(c) To find an antiderivative of $|x|$, we use a strategy we've used before with the absolute value function: we think about it piecewise. Find an antiderivative of $|x|$ on $(0, \infty)$. Then find an antiderivative of $|x|$ on $(-\infty, 0)$.

Solution

On $(0, \infty)$, $|x|$ is just x, so $\frac{1}{2}x^2$ is one antiderivative. On $(-\infty, 0)$, $|x|$ is equal to $-x$, so $-\frac{1}{2}x^2$ is an antiderivative.

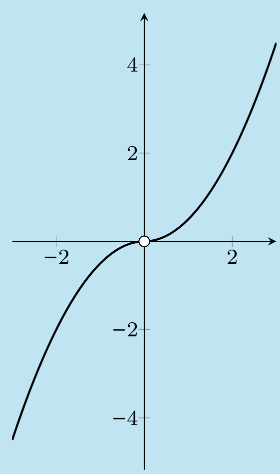
(d) Fill in the blanks in the following statement (by putting together your answers from the previous part), and sketch a graph of the function $F(x)$:

One antiderivative of $|x|$ is

$$F(x) = \begin{cases} \underline{\hspace{3cm}} & \text{if } x > 0 \\ \underline{\hspace{3cm}} & \text{if } x < 0 \\ \underline{\hspace{3cm}} & \text{if } x = 0 \end{cases}$$

Solution

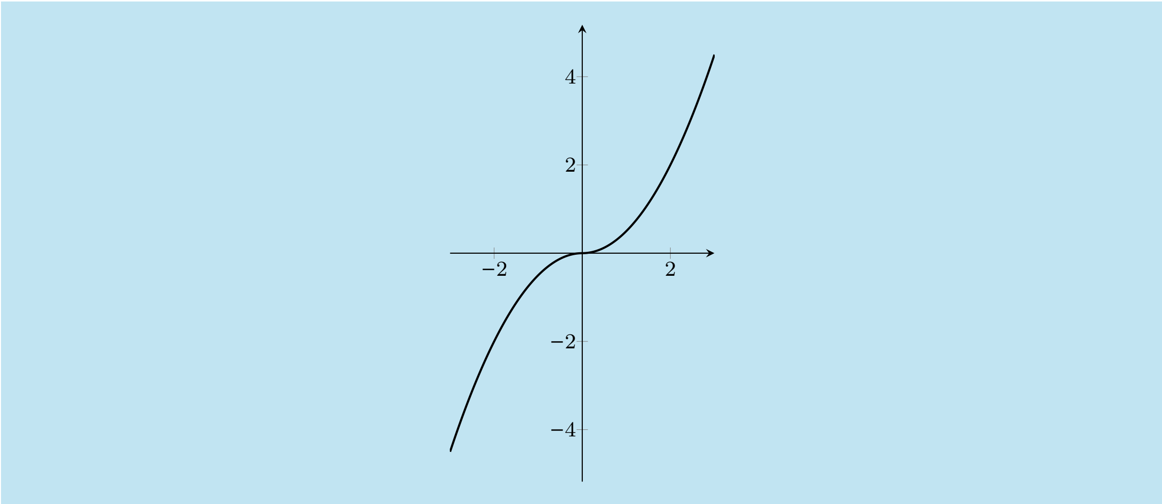
We already saw that $\frac{1}{2}x^2$ is an antiderivative of $|x|$ on $(0, \infty)$ and that $-\frac{1}{2}x^2$ is an antiderivative of $|x|$ on $(-\infty, 0)$. Graphed together, these look like:



However, the function we just graphed is not differentiable at 0 (but the function F we're trying to think of must be differentiable everywhere because its derivative, $|x|$, is defined everywhere). So, we need to define $F(0) = 0$. Putting this together, one antiderivative of $|x|$ is

$$F(x) = \begin{cases} \frac{1}{2}x^2 & \text{if } x > 0 \\ -\frac{1}{2}x^2 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \end{cases}$$

(You could also add a constant to this.) The graph of F looks like this:


4
2
−2
2
−2
−4